Evaluating reliability and explainability in seismic hazard forecasting for underground mine monitoring

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Research code: https://github.com/junjie-yang11/seismic-bump-forecasting

Abstract

Test-cohort composition materially changes the apparent validation gap in seismic hazard forecasting. We compare logistic regression (LR), bagged CART and training-tuned XGBoost on 2,063 common test rows from the 2,578-row UCI Seismic Bumps mirror. Record order is a temporal proxy because timestamps and longwall identifiers are absent. Matching rows reduces the apparent random-versus-record-order average-precision gap by 89.7 percent for LR and 86.9 percent for CART. Residual differences are 0.0114 (95 percent interval [-0.0110, 0.0243]) and 0.0188 ([0.0064, 0.0323]); the comparison does not isolate leakage or drift. Training-prior correction improves LR expected calibration error from 0.2904 to 0.0271 and Brier score from 0.1462 to 0.0439. Corrected LR Brier skill is 0.0342 against the historical training-prior reference and -0.0751 against the retrospective test-prevalence reference. Nine prespecified feature combinations and exact TreeSHAP distinguish fitted-model feature use from incremental predictive value. Energy, low-energy event counts and pulse activity repeatedly contribute to XGBoost predictions. At 10 and 20 percent training reference budgets, supplementary holdout XGBoost produces 30 and 81 alerts, detecting 3 and 6 of 26 hazardous shifts. Matched cohorts, explicit probability references and detection counts provide a traceable basis for interpreting forecast performance and inspection demand. Intervals condition on fixed predictions.

Keywords: mining engineering; seismic hazard forecasting; engineering feature ablation; TreeSHAP; explanation stability; warning budget

1 Introduction

Underground mine monitoring records rock-mass activity to support the assessment of hazardous conditions. Interpreting a forecast for engineering review requires evidence about its performance on later shifts, the monitoring signals it uses and the inspections its warnings would trigger. These questions connect model evaluation to detected hazards, missed hazardous shifts and false-alert workload.

The UCI Seismic Bumps data provide a defined next-shift forecasting task: each record summarizes an eight-hour shift, and the target indicates whether the following shift contains a seismic event above 10,000 J [1]. Existing seismic and seismoacoustic hazard ratings appear alongside energy, pulse and event-count measurements. The data therefore support comparisons of recorded assessments with measured signals. The target is next-shift high-energy seismic occurrence; confirmed rockburst accidents are outside its label definition.

Rule-based mining-hazard prediction predates this study. Sikora and Wrobel [9] evaluated shift and hourly forecasts using longwall-specific monitoring data and also transferred rules between working faces. Their hazard definition and cohorts differ from the public UCI task, so their classification results provide engineering context rather than a numerical benchmark for our average precision.

Janusz et al. [12] compared expert and sensor information and investigated cold start at new mining sites. Site adaptation requires identified locations and local histories; our common-row evaluation addresses protocol comparisons within an anonymous record sequence. The microseismic review by Anikiev et al. [13] distinguishes detection and location tasks from forecasting. This study uses shift summaries, not waveform detection or event-location models.

Validation protocols can change both the training history and the test population [3,4]. Comparing their scores on different shifts can therefore obscure the source of an apparent performance gap. A common test cohort makes that comparison interpretable, while probability assessment determines what the score scale conveys. Feature ablation, model explanations and frozen thresholds then connect the monitoring inputs to predictive contribution and warning workload.

This study connects three engineering questions: how cohort composition changes the validation comparison, whether probability correction improves assessment, and how monitoring signals contribute to predictions and warning workload. Common test rows reveal how cohort composition changes the apparent protocol gap. Training-local correction and explicit probability references separate calibration improvement from baseline-relative skill. TreeSHAP and nine prespecified, refitted feature combinations distinguish the inputs used by a fitted model from their incremental predictive value. Frozen reference budgets then express the forecasts as detections, missed hazardous shifts and inspection workload. These empirical findings use established learning and explanation methods [6,7], saved row-level evidence and training-only selection.

2 Data and engineering feature hypotheses

2.1 Dataset and provenance

The original UCI dataset contains 2,584 records from two longwalls of a Polish coal mine, with 170 positive target records [1]. The positive count refers to labelled shift records, rather than a catalogue of 170 independently identified seismic events. The CSV mirror retains 2,578 records [2]. A direct ARFF-to-CSV audit confirms first-occurrence deduplication without reordering: six excluded original rows, 90, 91, 973, 974, 1018 and 1019, are exact duplicate negatives. All positive records remain. The content-identical first-occurrence records retained for these rows have original IDs 88, 89, 971, 972, 1016 and 1017, respectively. These IDs are one-based data-row numbers excluding the ARFF header.

Row order is used as the temporal proxy because explicit timestamps and longwall identifiers are absent. The initial training block contains 15.92 percent positive records, compared with 6.98, 1.94, 4.65, 3.49 percent in the four subsequent test phases. Exact phase counts and prevalence are provided in Table 7. The full dataset prevalence is 6.59 percent, whereas the common test cohort has 88 positives among 2,063 records, or 4.27 percent. These differences motivate both matched evaluation and phase-specific analysis.

2.2 Engineering feature groups

Table 1 partitions the existing 17-column design into four engineering groups. The three hazard-rating variables encode recorded assessments, while the geophone and seismic-activity groups contain monitoring measurements. Shift type describes preparation or coal-getting activity. These groups define comparisons between existing hazard assessments and the measured signals available to a forecasting model.

Table 1. Engineering feature groups and the predictive hypotheses examined.

Group	Variables	Question
Hazard ratings	seismic, seismoacoustic, ghazard	How much information is already represented by recorded assessments?
Geophone	genergy, gpuls, gdenergy, gdpuls	Do energy, pulse activity and deviations add information beyond ratings?
Seismic activity	nbumps2 to nbumps89, energy, maxenergy	Do event counts by energy band and released-energy measures add information?
Operation	shift	Does shift type provide predictive context?

Categorical fields retain the existing ordinal encoding. The total count nbumps is excluded from the retained 17-column design, which includes the energy-band counts. Constant energy-band columns remain in the audited design. LR uses constant-safe training standardization and L2 regularization. The design uses the supplied shift summaries; constructing cross-shift lag features would require known longwall continuity.

3 Methods

3.1 Common test cohort and model comparison

Record-order evaluation divides the mirror into five consecutive blocks. Four expanding-window fits train on earlier blocks and predict the next block; the first block supplies training only. Every model and feature combination therefore evaluates the same 2,063 rows. A supplementary holdout trains on the first 70 percent and evaluates the last 774 records, containing 26 positive targets. This holdout was inspected in the earlier project, so it is reported as descriptive supplementary evidence and is not used to select a feature set, model or budget.

LR retains L2 penalty 1.0, an unpenalized intercept and balanced class weights. Bagged CART retains 60 ordinary bootstrap trees, maximum depth 6, minimum leaf size 20, unweighted Gini and forest seed 7. Every bootstrap draw is retained; single-class draws form constant leaf trees. Each feature supplies up to 12 split thresholds at quantiles 0.05 to 0.95. XGBoost uses unweighted binary logistic loss and histogram trees, learning rate 0.05, minimum child weight 10 and L2 penalty 5.0, with full row and column sampling, seed 7 and two CPU threads [6]. Its fixed candidate grid combines depth 2 or 3 with 80 or 160 boosting rounds.

Each XGBoost search uses only its supplied training rows: the final 20 percent forms an inner reference, and candidates are fitted on the preceding 80 percent. Selection maximizes tied-score average precision (AP), with negative Brier score as the predefined fallback for a zero-positive reference; the first grid entry wins exact ties. The selected model is refitted on the entire supplied training set. Threshold-reference models conduct their own search within the earlier inner-fit prefix. For feature ablations, full-feature training selects parameters once per fold and the same choices are used for all nine feature combinations.

The random protocol uses ten stratified folds and five seeds, 0 to 4. XGBoost selects parameters through a stratified inner split within each outer training fold. Random predictions are restricted to the record-order test cohort before calculating comparative metrics. Reported random AP is the mean of five seed-specific AP values, rather than AP after averaging scores. LR and CART retain fixed settings; XGBoost is training-tuned. Comparisons characterize these stated workflows rather than equally extensive searches for every model.

3.2 Feature comparisons and conditional uncertainty

The prespecified feature sets are the full design, each group alone and the design with each group removed. They are all reported, without choosing a final feature set from test outcomes. The full-versus-ratings-only contrast addresses the additional predictive information in monitoring measurements and shift type. Group-removal contrasts assess predictive dependence on information groups in the presence of correlated alternatives.

AP is the main ranking measure and groups tied scores before integration. ROC-AUC, Brier score and ten-quantile-bin expected calibration error (ECE) provide complementary discrimination and probability assessments. LR probability assessment uses each outer training fold's unweighted prior to correct its

balanced-weight output; CART and XGBoost are assessed as fitted. No test-label prior is used in forecasting or calibration.

Paired moving-block intervals use 2,000 replicates, block length 32 and RNG seed 20261002. Contiguous blocks are sampled separately within each test phase, preserving phase sizes and applying identical sampled rows to every compared score vector. Intervals are the 2.5th and 97.5th percentiles of paired AP differences and condition on fixed predictions. Random-protocol differences average the five seed-specific AP values within each replicate. The earlier LR/CART protocol analysis also checks lengths 16 and 64. The intervals are descriptive, unadjusted for multiple feature contrasts, and exclude model-refit and parameter-selection uncertainty [5].

3.3 Explanations and phase stability

XGBoost models with all features explain their own outer test predictions through native exact TreeSHAP. Training leaf covers provide its tree-path-dependent reference [7,8]. Each row has 17 feature contributions and a bias term. Their sum reconstructs the raw log-odds margin; applying the logistic function reconstructs the predicted probability. Contributions are additive on the model's log-odds scale and describe fitted predictive associations.

Global importance is mean absolute contribution on the common test cohort. Each phase also has its own importance ranks. Pairwise Spearman correlations compare the ranks of all 17 features, with average ranks for ties; top-five Jaccard overlap measures agreement in leading features. A fixed feature-name order resolves ties at the top-five boundary. Value-contribution Spearman associations and scatter plots describe how observed values relate to contributions. Together, the rank and overlap measures characterize recurring feature use and changes among leading inputs.

Illustrative cases use the earliest record in each true-positive, false-positive and false-negative category under the predeclared 10 percent training reference budget. Case selection is a post-evaluation explanation step, independent of model and threshold selection.

3.4 Training reference budgets and warning outcomes

The reference budgets are fixed at 1, 5, 10 and 20 percent. For each outer fold, an inner model predicts the last 20 percent of the training history. The threshold excludes the boundary score and all tied scores together, so at most floor(reference size times budget) reference records are flagged. After outer refitting, the numerical threshold is frozen and evaluated on future test records. All budgets use the same model scores; no test labels enter threshold selection.

For LR, both threshold selection and alert classification use raw weighted-model scores. Prior-corrected probabilities are used separately for probability assessment. The inner and outer models are distinct fits, so refitting can alter raw-score distributions and the transferred alert rate even though the probability scales are not mixed. A fixed-inner-model comparison at the same cutoff would quantify the difference between these workflows on identical test rows; it would not identify a causal effect of refitting.

Reference budgets are experimental workload scenarios, not established operating limits for a mine. The future actual alert rate is measured separately because refitting and changing score distributions can alter it. Warning outcomes include true detections, false alerts, missed hazardous shifts, precision and recall. The target and available labels measure shift-level detections, not event counts, spatial warning coverage or avoided accidents.

4 Predictive contribution of monitoring information

4.1 Cohort sensitivity and matched model comparison

Table 2 compares validation protocols on identical test records. Full-feature XGBoost record-order AP is 0.0862, compared with 0.0838 for LR and 0.0784 for CART. The XGBoost-minus-LR paired difference is 0.0024 with interval [-0.0290, 0.0291], and the XGBoost-minus-CART difference is 0.0079 with interval [-0.0019, 0.0267]. These paired comparisons establish the model context for the monitoring-information and warning-policy analyses below.

Table 2. AP on identical 2,063 test rows and paired random-minus-record-order intervals; random values average five seeds.

Model	Random AP	Record-order AP	Difference	95% interval
LR	0.0953	0.0838	0.0114	[-0.0110, 0.0243]
CART	0.0972	0.0784	0.0188	[0.0064, 0.0323]
XGBoost	0.0995	0.0862	0.0133	[0.0004, 0.0256]

Table 2 note. LR uses balanced class weights, bagged CART is unweighted, and XGBoost is unweighted with an additional training-only parameter search. This is a comparison of stated workflows with asymmetric weighting and tuning. The CART and XGBoost protocol intervals exclude zero; the LR interval includes zero.

Matching test rows reduces the original different-cohort AP gap by 89.7 percent for LR and 86.9 percent for CART. The size of this change establishes test-cohort composition as a material part of protocol comparison. It measures evaluation-population sensitivity; the remaining differences combine training history, training size and, for XGBoost, training-selected configurations, so the reduction is not a causal decomposition of leakage or drift.

4.2 Engineering feature comparisons

Table 3 reports every prespecified feature set. Hazard ratings alone yield AP 0.0596, 0.0738 and 0.0766 for LR, CART and XGBoost. The full-feature gains are 0.0243, 0.0046 and 0.0096, respectively (Table 4). Paired intervals in Table 4 include zero for all three gains. The comparisons quantify the observed contribution of measured signals and its conditional uncertainty.

Table 3. Engineering feature ablations on the common record-order cohort; entries are AP. All three models use the same columns per row.

Feature set	Columns	LR	CART	XGBoost
All features	17	0.0838	0.0784	0.0862
Hazard ratings only	3	0.0596	0.0738	0.0766
Without hazard ratings	14	0.0900	0.0766	0.0856
Geophone only	4	0.0792	0.1000	0.1025
Without geophone	13	0.0734	0.0794	0.0828
Seismic activity only	9	0.0751	0.0761	0.0785
Without seismic activity	8	0.1020	0.0952	0.1292
Shift type only	1	0.0504	0.0824	0.0824
Without shift type	16	0.0837	0.0778	0.0862

Table 4. Full-feature AP minus hazard-ratings-only AP with conditional paired 95 percent intervals.

Model	Incremental AP	95% interval
LR	0.0243	[-0.0039, 0.0619]
CART	0.0046	[-0.0252, 0.0264]
XGBoost	0.0096	[-0.0148, 0.0413]

The geophone-only XGBoost achieves AP 0.1025, while removal of the seismic-activity group gives 0.1292. The latter differs from the full model by 0.0430 with interval [-0.0034, 0.1067]. Together with the LR and CART contrasts, these results show that predictive use of a monitoring group depends on the other available inputs and fitted model. The comparisons hold XGBoost settings fixed within each fold, assessing the sensitivity of that design to information sources. Redundancy, estimation variability and changing phase distributions are candidate explanations for the observed patterns; they are not separately identified here.

4.3 Calibration improvement and reference choice

Training-prior correction improves LR ECE from 0.2904 to 0.0271 and Brier score from 0.1462 to 0.0439, providing a direct adjustment to the probability scale using historical labels. Corrected LR, raw CART and raw XGBoost Brier scores are 0.0439, 0.0433 and 0.0429. Corrected LR Brier skill against a retrospective evaluation-prevalence constant is -0.0751. The constant is an oracle reference used for assessment, with no role in generating forecasts. Table 5 also compares each model with a deployable historical reference: every test row receives the unweighted prevalence of its outer training fold. Corrected LR has Brier skill 0.0342 against this historical reference. ECE improvement compares LR before and after correction, while skill compares its squared error with a stated reference; the two assessments therefore answer different questions.

Table 5. Probability assessment on the common test cohort. LR uses corrected probabilities; CART and XGBoost use raw probabilities. Historical references are frozen within training; oracle references use the evaluation prevalence for retrospective assessment.

Model	Model Brier	Historical Brier	Historical skill	Oracle Brier	Oracle skill
LR	0.0439	0.0455	0.0342	0.0408	-0.0751
CART	0.0433	0.0455	0.0472	0.0408	-0.0607
XGBoost	0.0429	0.0455	0.0573	0.0408	-0.0494

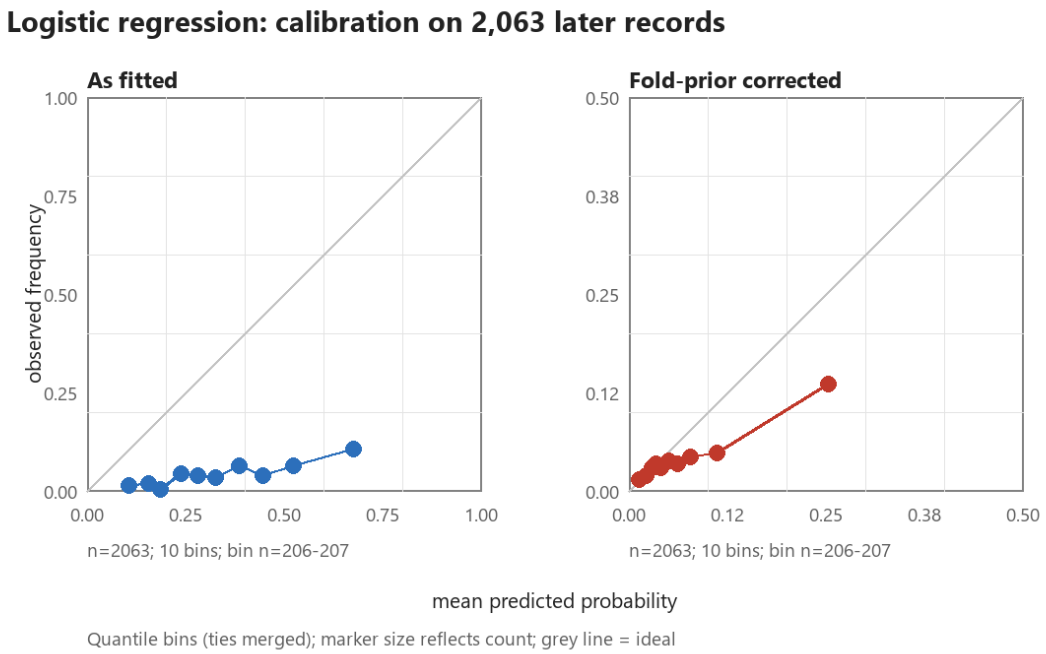


Figure 1. LR calibration on the common test cohort. Each panel contains 10 quantile bins of 206 or 207 observations; ties stay together, marker sizes encode counts, and panel axes differ.

5 Model explanations and phase stability

5.1 Global and stage-specific feature use

Total seismic energy, low-energy event count nbumps2 and pulse count gpuls have the largest mean absolute XGBoost contributions. These are observable indicators of seismic and geophone activity; their importance describes how the fitted models use monitoring information. Table 6 reports the leading contributions and value associations. Read alongside the ablations, these explanations distinguish a model's reliance on a signal from the incremental value of its monitoring group.

Table 6. Leading full-feature XGBoost contributions on 2,063 test rows. Magnitudes are in raw log-odds units; correlations are descriptive value-contribution associations.

Feature	Mean absolute contribution	Value association
energy	0.5642	0.6854
nbumps2	0.3902	0.7179
gpuls	0.3523	0.7668
genergy	0.1466	0.0148
seismic	0.1391	0.8297

The two largest contributions, energy and nbumps2, belong to the seismic-activity group. Removing that group and refitting gives pooled XGBoost AP 0.1292 versus 0.0862 for the full design, a difference of 0.0430 with paired interval [-0.0034, 0.1067]. The interval includes zero. SHAP attributes the predictions of the full fitted model; ablation evaluates a newly fitted model with different inputs. Model reliance and incremental predictive value are distinct: influential signals need not improve the performance of a refitted model under the tested configuration. The interval leaves the direction of the group-removal gain unresolved. Correlated inputs, fitted interactions and sampling variability are plausible contributors. Appendix Table A2 reports all feature sets by phase; for XGBoost, without-

seismic-minus-full AP differences are 0.1058, 0.0124, -0.0031, 0.0049 across phases 1 to 4. These phase patterns accompany the pooled comparison rather than identify its cause.

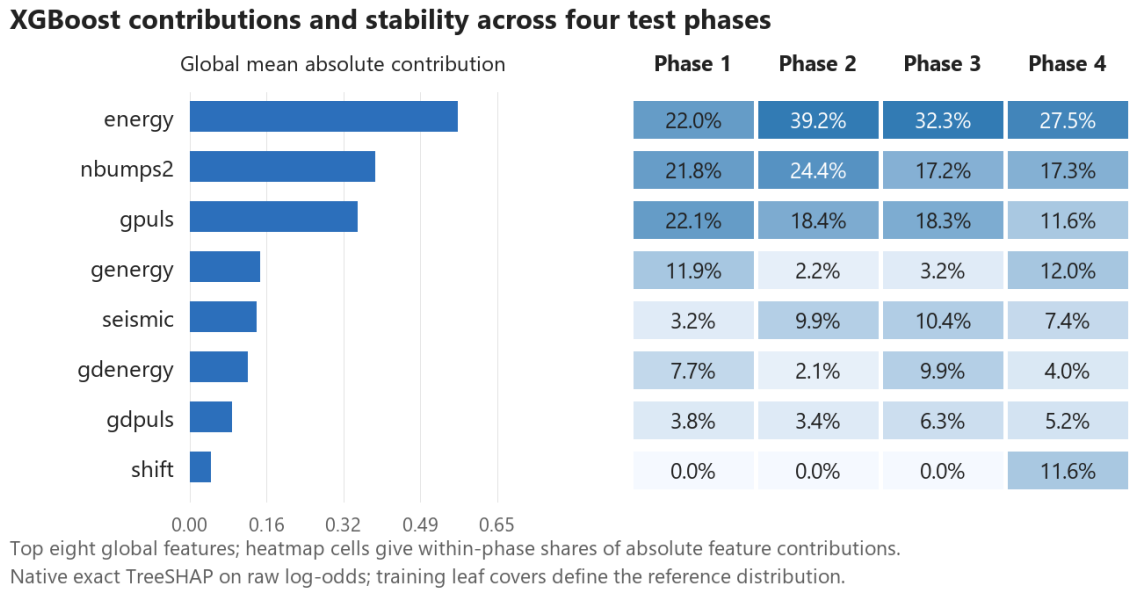


Figure 2. Global importance and phase shares for the eight globally leading features. Each cell is that feature's mean absolute contribution divided by the sum over all 17 features in the phase.

Across the six phase pairs, importance-rank correlations range from 0.7122 to 0.9412 and top-five Jaccard overlap ranges from 0.4286 to 0.6667. This combination shows recurring feature use alongside changing membership of the most influential group. Excluding the globally constant nbumps6, nbumps7 and nbumps89 leaves one fixed 14-feature universe for every phase and gives correlations from 0.6570 to 0.9342. Ranks are recomputed using average ranks on each fixed universe; alphabetical ordering resolves only top-five membership ties. Top-five overlap is unchanged by excluding these constants. Appendix Table A3 reports every phase pair under both universes.

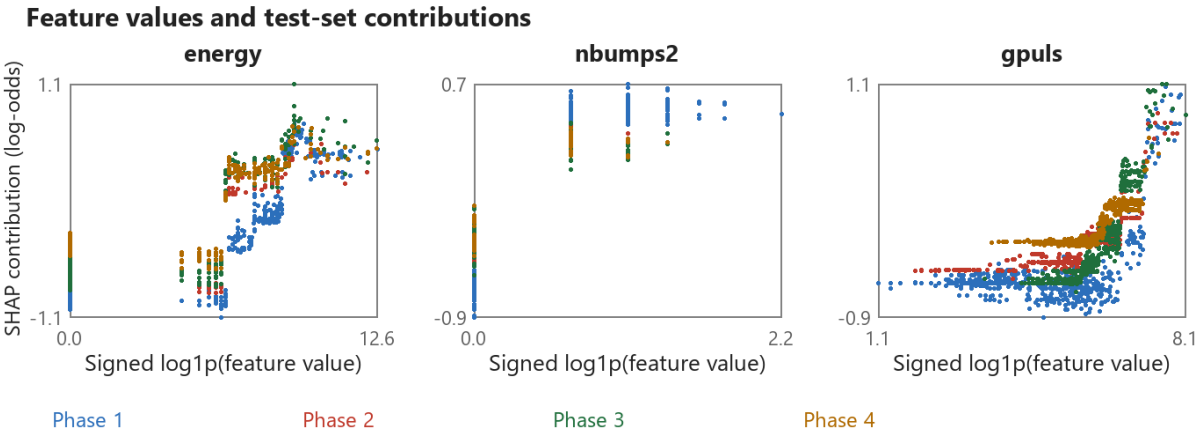


Figure 3. Values and TreeSHAP contributions for the three globally leading features. Colours identify test phases; horizontal values use signed log1p for display, and vertical values are raw log-odds contributions.

The scatter distributions expose phase-dependent and nonlinear feature use. A high global importance can coexist with a weak global monotonic association: genergy, for example, has a descriptive value-contribution correlation of 0.0148. The phase-coloured patterns show how a measurement's contribution varies with the other inputs and the fitted model. For engineering review, the contribution describes the signal in its prediction context.

5.2 Phase outcomes and illustrative warning cases

Table 7 connects explanation changes to the phase outcomes. Positive prevalence and full-model AP vary across the record sequence. Computing AP within each block separates its observed discrimination from the cross-model score-scale differences that can affect pooled AP. Together, Tables 6 and 7 show why monitoring review benefits from examining feature use and phase outcomes side by side.

Table 7. Four common test phases with positive counts, prevalence and within-phase full-feature AP.

Phase	Rows	Positives	Prevalence	LR AP	CART AP	XGBoost AP
1	516	36	0.0698	0.1360	0.1136	0.1341
2	515	10	0.0194	0.0204	0.0274	0.0284
3	516	24	0.0465	0.0873	0.0632	0.0731
4	516	18	0.0349	0.1142	0.0781	0.0618

The predetermined holdout cases are shown in Figure 4. Each case decomposes the bias and all feature contributions to reconstruct the model margin. The five largest absolute contributions are displayed individually, with remaining contributions combined as Other features. The three outcomes show how combinations of measured signals place different records above or below the same frozen cutoff. Signed contributions locate each model adjustment relative to its baseline.

The TP, FP and FN categories depend on predicted alerts and observed labels. The first row in each category supplies a deterministic illustration with limited discretionary case selection.

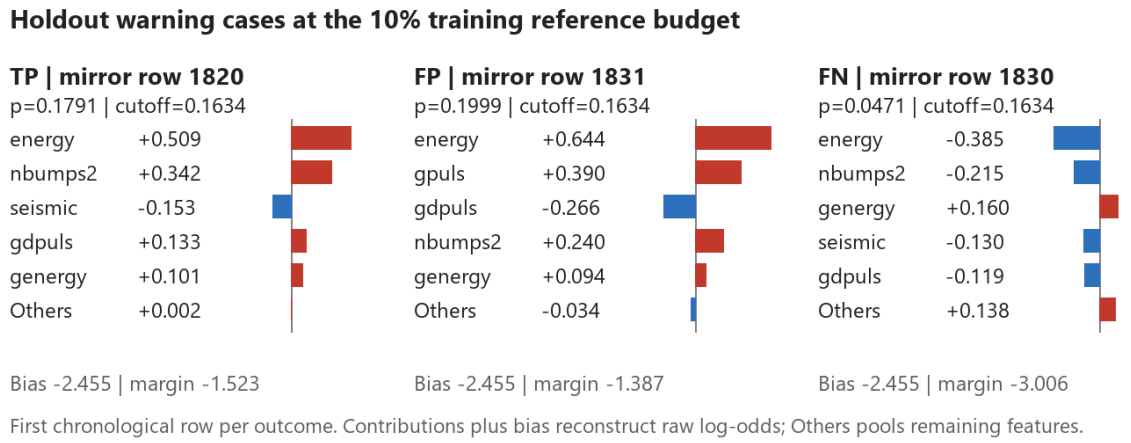


Figure 4. Earliest holdout true-positive, false-positive and false-negative cases under the 10 percent reference budget. IDs are one-based mirror rows; raw log-odds contributions, including Other features, sum with the bias to the margin. Bars use a common symmetric scale.

6 Warning budgets and inspection workload

Changing the reference budget changes the frozen threshold while leaving fitted scores unchanged. In the common record-order cohort, LR, CART and XGBoost actual alert rates at the 10 percent reference budget are 0.1609, 0.0785 and 0.0999. This provides a direct comparison between intended historical workload and its transferred operating outcome.

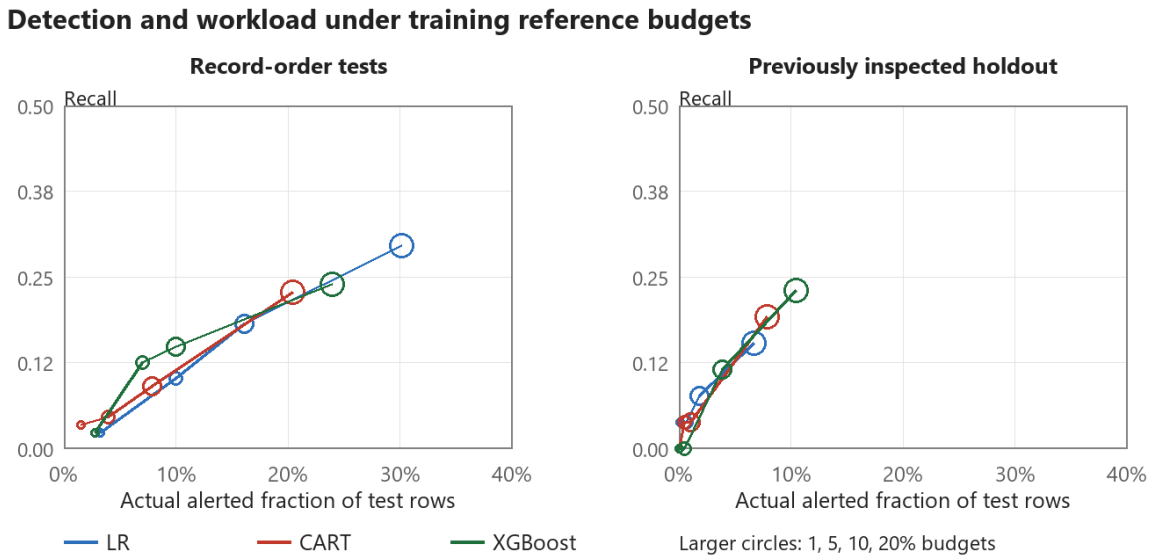


Figure 5. Recall versus actual test alert fraction under the four fixed training reference budgets. Colours identify models and increasing circle sizes identify 1, 5, 10 and 20 percent budgets.

Table 8 reports the supplementary holdout outcomes for every model and budget. For XGBoost, increasing the reference budget from 10 to 20 percent increases alerts from 30 to 81 and detections from 3 to 6, while missed hazardous shifts change from 23 to 20. The corresponding actual alert rates are 0.0388 and 0.1047. Reporting both detection counts and actual alert rates makes the workload change measurable when a historical threshold is transferred to later records.

Table 8. All holdout warning policies on 774 records with 26 positive targets. Budget is the training reference fraction; alert rate is measured on holdout rows. Precision is reported as zero when no alerts are issued.

Model	Budget	Alerts	Detected	False alerts	Missed	Alert rate	Recall	Precision
LR	1%	1	1	0	25	0.0013	0.0385	1.0000
LR	5%	5	1	4	25	0.0065	0.0385	0.2000
LR	10%	14	2	12	24	0.0181	0.0769	0.1429
LR	20%	52	4	48	22	0.0672	0.1538	0.0769
CART	1%	0	0	0	26	0.0000	0.0000	0.0000
CART	5%	4	1	3	25	0.0052	0.0385	0.2500
CART	10%	8	1	7	25	0.0103	0.0385	0.1250
CART	20%	61	5	56	21	0.0788	0.1923	0.0820
XGBoost	1%	0	0	0	26	0.0000	0.0000	0.0000
XGBoost	5%	4	0	4	26	0.0052	0.0000	0.0000
XGBoost	10%	30	3	27	23	0.0388	0.1154	0.1000
XGBoost	20%	81	6	75	20	0.1047	0.2308	0.0741

At the 1 percent reference budget, holdout LR issues one alert and detects one hazardous shift, while CART and XGBoost issue no alerts. Detection and missed-shift counts make these sparse outcomes interpretable alongside precision. Higher budgets increase detections together with additional false-alert

workload. The policy curves describe the observed detection–workload tradeoff. Choosing an operating budget would also require defined inspection capacity and consequences of missed hazards.

7 Discussion

7.1 Connecting monitoring evidence to warning decisions

Matching test rows sharply reduces the LR and CART validation gaps, showing that the evaluation cohort materially affects their apparent size. The residual gaps compare workflows that still differ in training history and size; they do not isolate a leakage effect. Probability assessment adds a second distinction. Training-prior correction improves LR calibration, but corrected probabilities have positive Brier skill against the historical reference and negative skill against the retrospective test-prevalence reference. Improvement over the raw model and advantage over a reference are separate findings.

SHAP and ablation answer different engineering questions. Energy and low-energy event counts strongly influence the fitted XGBoost predictions, while refitting without their group yields a positive pooled AP difference whose interval includes zero. Their influence describes the fitted prediction mechanism; the ablation tests the performance of a different fitted model. Reading these results together prevents global importance from becoming a claim of incremental value. The warning-budget results extend this interpretation to decisions: additional detections come with additional false alerts, and a historical reference budget can produce a different alert rate on later records.

7.2 Scope and prospective validation

Changing a probability scale and changing a decision rule address different engineering questions. Cost-sensitive decision theory [11] links probabilities to error costs; here historical reference scores supply empirical cutoffs. Prior-only distribution change [10] is a specific assumption rather than a diagnosis of the observed protocol gap. The companion decision study evaluates loss, capacity and refitting at frozen thresholds; it builds on this forecast evidence without treating calibration gains as evidence of operational warning value.

Threshold-choice theory [14] places the scoring model, decision rule and operating conditions in the same loss framework. A reference budget selects a historical score cutoff; carrying that number forward does not enforce a future alert rate. Stage two makes fixed-model transfer its primary workflow and compares refitting separately. Its frozen-rule cost boundaries retain the historical selection cost while varying evaluation cost, and its prevalence scenarios retain class-conditional rates while varying class proportions. Precision changes under the latter assumption need not indicate changed ranking [15]. These decision calculations complement the present AP, probability-reference and feature-use evidence.

These results concern the audited mirror under a record-order forecasting assumption. Timestamps, longwall identifiers and event locations are needed for direct temporal, site-specific and spatial validation. Repeated measurements can correspond to distinct shifts; the provenance audit establishes the mirror transformation rather than the operational validity of deduplication. Native TreeSHAP uses training leaf covers, and correlated inputs affect the allocation of contributions. Signed contributions explain how the fitted model uses a variable, not the causal effect of changing that variable. Phase stability is conditional on the fitted models and observed test distributions.

The feature comparisons are prespecified descriptive contrasts with unadjusted fixed-prediction intervals. A maximum AP among these feature sets is not treated as a validated model-selection result. The already inspected holdout supplies additional warning evidence but does not constitute an untouched confirmatory test. The cold-start question in [12] concerns adapting to a new site with limited local history. Our record blocks assess evaluation and cutoff transfer, without identifying site changes. A

subsequent study should preregister its feature and alert-policy choices, then test them on independent timestamped working-face data with defined inspection actions.

8 Conclusions

This study evaluates seismic forecasts through matched test records, probability references and monitoring-feature comparisons. Matching rows reduces the apparent protocol gap without identifying a causal share of leakage or drift. Training-prior correction improves LR calibration, while baseline-relative Brier skill depends on the stated reference. TreeSHAP describes fitted-model feature use; refitted ablations assess incremental predictive value. Frozen-budget outcomes connect these forecasts to detected, missed and falsely alerted shifts. The workflow links saved predictions to probability assessment, feature comparisons and inspection demand, making the scope of each engineering claim explicit.

Data and computational reproducibility

Source data are available from UCI [1] and the CSV mirror [2]. Saved evidence includes predictions for every feature set, scores from inner reference sets, fold audits, predictions for candidate selection and manifests of row indices. It also retains bootstrap replicates, five random seeds, native XGBoost models, TreeSHAP contributions for each row and the predetermined case records. Result tables are generated from saved evidence. Source hashes and duplicate row mappings preserve the data version. Appendix Table A1 maps feature names to UCI definitions and the implemented engineering groups.

The extended analysis uses Python 3.12.14, NumPy 2.3.5, pandas 3.0.1, XGBoost 3.1.3 and SciPy 1.18.1. The legacy LR/CART baselines use the original verified Python 3.7 environment and retain their saved full-model predictions. Run commands from the repository root. Use the recorded baseline environment for python -m phase1.run_experiments and python -m phase1.run_engineering. For the extension, install requirements/requirements-research.txt in a separate Python 3.12 environment, then run python -m phase1.run_research, python -m phase1.review_analysis and python -m phase1.verify_research --replay. In a separate document environment, python -m phase1.generate_report --docx generates the paper; pwsh -NoProfile -ExecutionPolicy Bypass -File phase1/export_report.ps1 exports PDF. Use python -m phase1.verify_results --reports to check results and report agreement.

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The author directed the research scope, protocol refinement and review priorities. ChatGPT and Codex substantially assisted implementation, execution, computational verification, reference-metadata checks and writing. The repository CONTRIBUTIONS.md separates author review from recorded computational checks and links project decisions to evidence.

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Appendix A Feature definitions and phase evidence

Table A1. Implemented feature groups and concise UCI variable definitions [1]. GMax is the most active geophone. Constant marks a numeric column with one value in the audited mirror.

Feature	Group	Definition	Encoding
seismic	Hazard ratings	Seismic-method shift hazard rating	Ordinal a-d
seismoacoustic	Hazard ratings	Seismoacoustic shift hazard rating	Ordinal a-d
ghazard	Hazard ratings	GMax geophone hazard rating	Ordinal a-d
genergy	Geophone	Previous-shift energy at GMax	Numeric
gpuls	Geophone	Previous-shift pulse count at GMax	Numeric
gdenergy	Geophone	Energy deviation from previous eight-shift mean	Numeric
gdpuls	Geophone	Pulse-count deviation from previous eight-shift mean	Numeric
shift	Operation	Coal-getting W or preparation N	Binary
energy	Seismic activity	Previous-shift total bump energy	Numeric
maxenergy	Seismic activity	Previous-shift maximum bump energy	Numeric
nbumps2	Seismic activity	Previous-shift bump count in [10^2, 10^3) J	Numeric
nbumps3	Seismic activity	Previous-shift bump count in [10^3, 10^4) J	Numeric
nbumps4	Seismic activity	Previous-shift bump count in [10^4, 10^5) J	Numeric
nbumps5	Seismic activity	Previous-shift bump count in [10^5, 10^6) J	Numeric
nbumps6	Seismic activity	Previous-shift bump count in [10^6, 10^7) J	Constant

Feature	Group	Definition	Encoding
nbumps7	Seismic activity	Previous-shift bump count in [10^7, 10^8) J	Constant
nbumps89	Seismic activity	Previous-shift bump count in [10^8, 10^10) J	Constant

The total count nbumps is excluded from the retained 17-column design, which includes the energy-band counts. Ordinal ratings and the binary shift type use the fixed mappings documented in the preprocessing code; scaling is fitted within training.

Table A2. Within-phase AP for every prespecified feature set. Phases use the same test rows across models and subsets. XGBoost subsets reuse the configuration selected on full-feature training data.

Model	Feature set	Phase 1	Phase 2	Phase 3	Phase 4
LR	All features	0.1360	0.0204	0.0873	0.1142
LR	Hazard ratings only	0.1300	0.0379	0.0474	0.0355
LR	Without hazard ratings	0.1298	0.0164	0.0911	0.1589
LR	Geophone only	0.1731	0.0390	0.0547	0.1103
LR	Without geophone	0.1177	0.0418	0.1224	0.0649
LR	Seismic activity only	0.0979	0.0194	0.1415	0.0768
LR	Without seismic activity	0.2310	0.0298	0.0666	0.0456
LR	Shift type only	0.1104	0.0303	0.0658	0.0392
LR	Without shift type	0.1340	0.0174	0.0874	0.1204
CART	All features	0.1136	0.0274	0.0632	0.0781
CART	Hazard ratings only	0.1171	0.0362	0.0476	0.0306
CART	Without hazard ratings	0.1111	0.0171	0.0698	0.0628
CART	Geophone only	0.2131	0.0360	0.0628	0.0461
CART	Without geophone	0.1100	0.0366	0.0800	0.0542
CART	Seismic activity only	0.1032	0.0194	0.1145	0.0514
CART	Without seismic activity	0.2119	0.0341	0.0618	0.0471
CART	Shift type only	0.1104	0.0303	0.0658	0.0392
CART	Without shift type	0.1136	0.0264	0.0623	0.0768
XGBoost	All features	0.1341	0.0284	0.0731	0.0618
XGBoost	Hazard ratings only	0.1171	0.0363	0.0465	0.0306
XGBoost	Without hazard ratings	0.1241	0.0215	0.0777	0.0701
XGBoost	Geophone only	0.1794	0.0349	0.0660	0.0791
XGBoost	Without geophone	0.1182	0.0351	0.0876	0.0594
XGBoost	Seismic activity only	0.1153	0.0194	0.1096	0.0599
XGBoost	Without seismic activity	0.2399	0.0408	0.0700	0.0667
XGBoost	Shift type only	0.1104	0.0303	0.0658	0.0392
XGBoost	Without shift type	0.1341	0.0284	0.0731	0.0627

Table A3. Phase-pair stability with a fixed universe of 17 columns and after excluding the three constant columns. Average ranks are used for Spearman; top-five overlap uses the stated alphabetical boundary rule.

Phase pair	Spearman 17	Spearman 14	Top-5 Jaccard 17	Top-5 Jaccard 14
1-2	0.8354	0.8023	0.4286	0.4286
1-3	0.8178	0.7932	0.6667	0.6667
1-4	0.7122	0.6570	0.6667	0.6667
2-3	0.9412	0.9342	0.6667	0.6667
2-4	0.8306	0.7879	0.4286	0.4286
3-4	0.8047	0.7277	0.4286	0.4286